



Methanol poisoning with bilateral basal ganglia necrosis and hemorrhage without visual impairment: A case report

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ABSTRACT

This manuscript describes an unusual case of a 15-year-old male who ingested alcohol fuel, resulting in severe methanol poisoning (blood concentration 175.71 mg/dL). Despite bilateral basal ganglia necrosis and hemorrhage confirmed by CT and MRI, the patient maintained normal vision. Immediate treatment with sodium bicarbonate, hemodialysis, methylprednisolone, folic acid, and vitamin B1 resulted in consciousness recovery within 24 h and normalization of metabolic parameters. Ophthalmologic examination revealed mild right optic disc edema but preserved visual acuity (20/20), normal visual fields, and color vision. At 12-week follow-up, neurological function continued to improve with no visual deterioration. This case demonstrates selective basal ganglia vulnerability in methanol toxicity with sparing of the visual system. Emergency physicians should conduct comprehensive neurological evaluations in methanol poisoning cases regardless of visual symptoms and consider investigating protective mechanisms against methanol-induced optic neuropathy.

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1. Introduction

Methanol poisoning causes metabolic acidosis and multi-organ damage via formic acid, a metabolite inhibiting mitochondrial function [1]. Classic presentations include visual impairment (from optic nerve toxicity), neurological deficits, and basal ganglia necrosis [2–4]. Over 20 % of severe cases develop combined vision loss and persistent neurological dysfunction, with basal ganglia lesions correlating with poor outcomes [5,6].

However, we describe an atypical case of methanol toxicity with isolated basal ganglia damage and intact visual function, challenging traditional pathophysiological models.

2. Case report

A 15-year-old male presented to a local hospital with nausea and vomiting of undetermined origin. With no prior medical history, he received intravenous fluids and symptomatic management. His condition deteriorated rapidly, progressing to unconsciousness and seizures. Two days later, he was transferred to our hospital in critical condition with a Glasgow Coma Scale score of 3 (E1V1M1). Vital signs included blood pressure 110/70 mmHg, heart rate 110 beats/min, respiratory rate 22

breaths/min, and temperature 36.8 °C. Neurological assessment revealed bilateral pupillary dilation with diminished light reflexes.

Laboratory analysis demonstrated severe metabolic acidosis (pH 7.048, base excess −24.13 mmol/L, bicarbonate 7.2 mmol/L), elevated anion gap (28 mEq/L), increased creatinine (1.4 mg/dL), and lactic acidosis (lactate 6.2 mmol/L). Neuroimaging revealed bilateral basal ganglia abnormalities. Computed tomography showed symmetric hypodensities predominantly affecting the putamen with hemorrhagic components. Magnetic resonance imaging confirmed these findings, demonstrating bilateral hyperintense signals on T2-weighted sequences with susceptibility artifacts consistent with hemorrhage (Fig. 1).

Based on clinical and radiographic findings, methanol poisoning was strongly suspected. Treatment included intravenous sodium bicarbonate, mechanical ventilation, hemodialysis, methylprednisolone, folic acid (1 mg daily), and vitamin B1 (100 mg daily). Toxicology confirmed methanol presence with initial blood concentration at 175.71 mg/dL—markedly elevated. After 48 h of therapy, methanol levels decreased to 1.59 mg/dL (Table 1).

The patient's neurological status improved dramatically after 24 h, with GCS advancing to 14 (E4V4M6). Upon regaining consciousness, he acknowledged consuming approximately 200 ml of alcohol-based fuel, containing 95 % ethanol with unspecified methanol content. Metabolic parameters normalized by the third day. Ophthalmologic evaluation revealed mild right optic disc edema while the left eye appeared normal. Optical coherence tomography demonstrated normal

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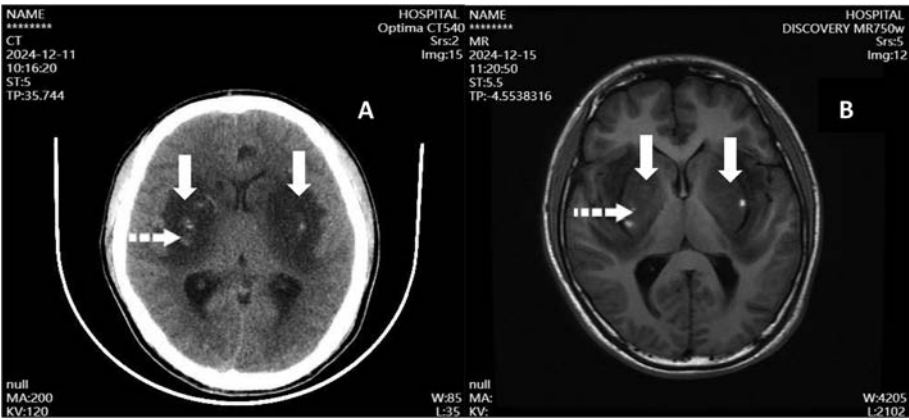


Fig. 1. Bilateral low-density lesions in the basal ganglia and corona radiata(A, full arrow), accompanied by minor hemorrhage(A, dashed arrow). Subsequent MRI revealed symmetrical abnormal signals in both cerebral hemispheres, indicating toxic encephalopathy with minor hemorrhage(B, full arrow).

retinal thickness. The patient maintained intact visual function with bilateral 20/20 visual acuity and normal visual field and color vision testing.

At 12-week follow-up, the patient showed continued neurological improvement with resolution of movement disorders and preserved visual function.

3. Discussion

Methanol poisoning typically presents with visual disturbances, metabolic acidosis, and neurological symptoms. However, our case presented with bilateral basal ganglia necrosis and hemorrhage with preserved visual function, challenging traditional understanding of methanol toxicity.

The susceptibility of the basal ganglia to formic acid may be related to its unique vascular structure and high metabolic demands. This region accounts for 7 % of the brain's glucose consumption and has distinctive venous drainage via the Rosenthal basal vein [13,14]. Formic acid inhibits cytochrome c oxidase a3, disrupting ATP synthesis, with greater impact in high-metabolism regions [15,16]. Post-mortem studies have

shown higher formic acid concentrations in the basal ganglia compared to other brain regions [17].

The preservation of visual function in our patient may be attributed to several protective factors. Ethanol's 10–20 times higher affinity for alcohol dehydrogenase (ADH) than methanol may have slowed methanol metabolism [9]. Retinal glial cells provide antioxidant defense through glutathione synthesis [9,18]. The optic nerve's stronger autoregulatory capacity may maintain better perfusion during acidosis [16].

Age-related neuroplasticity likely played a role, as adolescents have 3–5 times higher oligodendrocyte precursor cell activity than adults, potentially resisting axonal damage through remyelination [18]. Higher mitochondrial density in younger individuals may enhance hypoxia resistance [19].

Timing effects were also significant. Basal ganglia damage typically occurs 24–48 h post-poisoning, while optic nerve damage develops after 72 h [9,10]. Early ethanol ingestion naturally antagonized methanol effects, and hemodialysis at approximately, 48 h likely prevented optic nerve damage.

This case suggests differential sensitivities to methanol toxicity across brain regions and highlights the importance of comprehensive

Table 1
Clinical and laboratory parameters during hospital course.

Parameters	Reference interval	Admission	12 h	24 h	48 h	72 h
GCS Score	15	3	8	12	12	15
BP (mmHg)	90–140/60–90	95/52	99/65	108/62	115/66	127/74
Heart rate (beats/min)	60–100	112	124	98	85	68
Arterial Blood Gas						
PH	7.35–7.45	7.048	7.215	7.467	7.529	7.410
HCO3 [−] (mmol/L)	22–28	7.2	11.7	26.1	26	25.3
Base Excess (mmol/L)	−3 ~ +3	−24.13	−16.7	2.2	2	1.3
PaCO2 (mmHg)	35–45	15.3	22.4	36.1	29.2	42.2
Anion Gap (mmol/L)	8–16	28.1	23.6	11.5	12.1	8.7
Lactate (mmol/L)	0.5–2	12.0	8.5	2.5	1.5	1.6
Biochemistry						
Sodium (mmol/L)	135–145	136.6	133.5	139.6	140.3	131.6
Potassium (mmol/L)	3.5–5.5	5.48	4.42	3.11	3.10	3.95
Chloride (mmol/L)	96–106	99.6	105.4	105.9	107.5	100.9
ALT (U/L)	0–40	10.8	–	30	29.1	21.6
AST (U/L)	0–40	88.7	–	154.9	86.5	17.3
BUN (mmol/L)	2.7–7.0	7.31	–	2.65	–	8.83
Cr (μmol/L)	53–106	120.1	–	66.5	–	50.4
Toxicology Screen						
Blood methanol (mg/dL)	0–0.05	175.71*	–	–	1.59	0.02
Ophthalmologic Findings						
Visual acuity (Right)	20/20	–	–	20/20	20/20	20/20
Visual acuity (Left)	20/20	–	–	20/20	20/20	20/20
Pupillary light reflex	Normal	Normal	Normal	–	Normal	Normal
Papilledema	None	Slight	Minimal	–	None	None

neurological evaluation in methanol poisoning cases. Despite good initial visual recovery, we recommend ophthalmologic follow-up for at least 6 months, including OCT and visual evoked potential testing, to detect any delayed complications.

4. Conclusion

This case redefines the spectrum of methanol toxicity, demonstrating dissociated neurological and visual involvement. Comprehensive neuroimaging and aggressive early treatment are critical, irrespective of visual symptoms. Further research should explore protective mechanisms in the visual pathway and organ-specific vulnerability factors.

CRediT authorship contribution statement

Wei Pan: Writing – original draft, Formal analysis, Data curation. **Mei Liu:** Investigation, Data curation, Conceptualization. **Shuman Zhang:** Methodology, Formal analysis, Data curation. **Yonghong Wang:** Visualization, Validation, Methodology. **Xianyi Yang:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Consent to participate

Written informed consent was obtained from the patient's legal guardian.

Consent for publication

The patient's legal guardian provided written informed consent for the publication of this case report and accompanying images.

Ethics approval

This case report was approved by the Ethics Committee of Taihe Hospital, Hubei University of Medicine. Written informed consent was obtained from the patient's guardian for publication of this case report and accompanying images.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data generated or analyzed during this study are included in this published article.

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